Specific Heat Capacity

Exam code: 0625 & 0972

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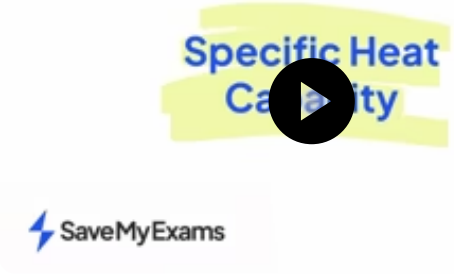

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Internal energy

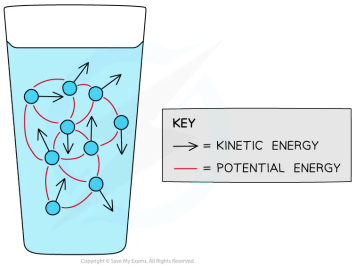
- A rise in the temperature of an object increases its **internal energy**
  - This can be thought of as due to an increase in the average speed of the particles
  - Increasing speed increases kinetic energy

Help me understand

- Internal energy is defined as:

The total energy stored inside a system by the particles that make up the system due to their motion and positions

- **Motion** of the particles affects their **kinetic energy**
- **Positions** of the particles relative to each other affects their **potential energy**
  - Together, these two make up the **internal energy** of the system



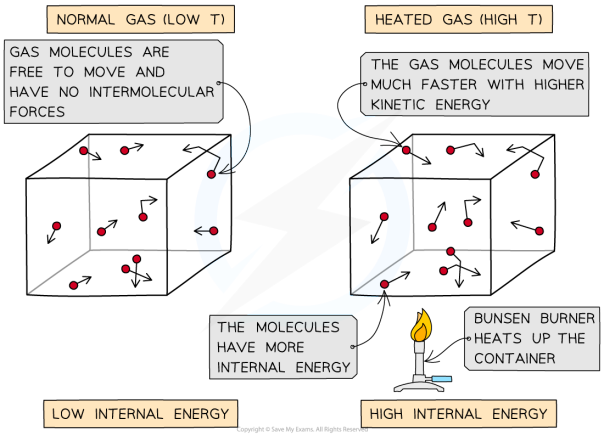
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Average kinetic energy

Extended tier only

- Heating a system changes a substance's internal energy by increasing the kinetic energy of its particles
  - The **temperature** of the material, therefore, is related to the **average kinetic energy** of the molecules

- An **increase** in **temperature** leads to an **increase** of the **average kinetic energy** of the particles in the substance
  - This also means that **internal energy increases,** as internal energy is the sum of all kinetic and potential energies



As the container heats up, the gas molecules move faster. Faster motion causes higher kinetic energy and therefore higher internal energy

Worked Example

What property of an object is a measure of the energy in the kinetic stores of its particles?

Answer:

- As temperature increases, so does the average kinetic energy of the particles
- This means **temperature** is a **measure** of the energy in the kinetic stores of the particles

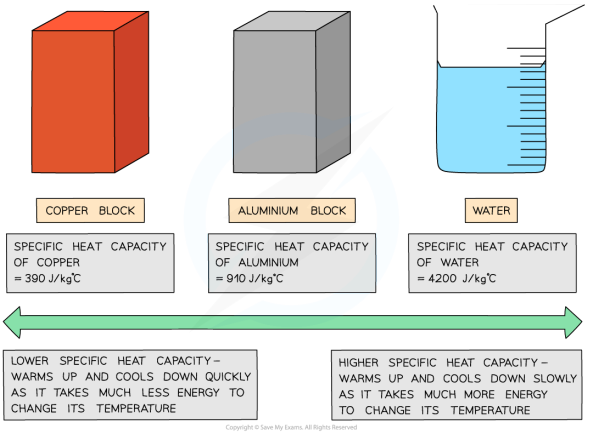
Specific heat capacity

- When heated, a substance's temperature can increase
- The amount by which the temperature increases depends on:
  - The **mass** of the substance heated
  - The **type** of material
  - The amount of **thermal energy** transferred in to the system

- The **specific heat capacity**, c, of a substance is defined as:

The energy required per unit mass per unit temperature increase

- In other words, this is the amount of energy required to raise the temperature of 1 kg of a substance by 1 °C
- Different substances have different specific heat capacities
  - If a substance has a **low** specific heat capacity, it heats up and cools down quickly (i.e. it takes less energy to change its temperature)
  - If a substance has a **high** specific heat capacity, it heats up and cools down slowly (i.e. it takes more energy to change its temperature)



Low vs high specific heat capacity

Calculating specific heat capacity

- The amount of energy needed to raise the temperature of a given mass by a given amount can be calculated using the equation:

$$c = \frac{\Delta E}{m \Delta \theta}$$

- Where:
  - ΔE = change in thermal energy, in joules (J)
  - m = mass, in kilograms (kg)
  - c = specific heat capacity, in joules per kilogram per degree Celsius (J/kg °C)
  - $\Delta \theta$ = change in temperature, in degrees Celsius (°C)

Worked Example

What unit is used to measure energy when calculating specific heat capacity?

A. kilograms

B. joules

C. joules per kilogram per degree Celsius

D. kelvin

Answer: B

- The question asks what unit is used to measure **energy** when calculating specific heat capacity
- The units of energy are **joules**
  - A common mistake is for students to see 'specific heat capacity' and state the units of that, not energy
  - Reading the question thoroughly and underlining key information avoids these mistakes

Worked Example

Water of mass 0.48 kg is increased in temperature by 0.7 °C. The specific heat capacity of water is 4200 J / kg °C.

Calculate the amount of thermal energy transferred to the water.

Answer:

Step 1: Write down the known quantities

- Mass, m = 0.48 kg
- Change in temperature, ΔT = 0.7 °C
- Specific heat capacity, c = 4200 J/kg °C

Step 2: Write down the relevant equation

$$\Delta E = mc\Delta\theta$$

Step 3: Calculate the thermal energy transferred by substituting in the values

$$\Delta E = (0.48) \times (4200) \times (0.7) = 1411.2$$

Step 4: Round the answer to 2 significant figures and include the units

$$\Delta E = 1400 \text{ J}$$

Examiner Tips and Tricks

While you must remember the equation for specific heat capacity, you will always be given the specific heat capacity of a substance so you do not need to memorise any values.

You can see this for yourself in your own kitchen at home. Metal pans, which have a relatively low specific heat capacity get very hot, very quickly when put on the hob. Add water to the pan, which has a relatively high specific heat capacity and the water will take much longer to heat up.

Notice the **units** of specific heat capacity:

joules *per* kilogram *per* degree Celsius : J / kg °C

'per' means 'divided by'. We say 'per' in front of every value that is being divided by, hence 'per kilogram per degree Celsius'

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